Deformation Cohomology of Twin Operations

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Abstract

We develop a complete deformation cohomology theory for *twin operations*, a group–action–based formalism introduced in Article I. Given a set E equipped with a binary operation $\alpha : E \times E \to E$ and a group G acting on E, we consider the full family of *twin operations*

$$\alpha_g(x, y) = g^{-1} \cdot \alpha(g \cdot x,\ g \cdot y), \qquad g \in G,$$

which forms an orbit of operations canonically identified with the homogeneous space $L\backslash G$ of *right* cosets, where $L = \{g \in G : \alpha_g = \alpha\}$ is the linear subgroup. This orbit perspective—viewing a symmetry as generating a *space* of operations rather than a single invariant one—is the key structural ingredient that makes deformation cohomology possible in this setting.

We show that this twin–space formalism admits a Hochschild-type cohomology. In particular: (i) $\mathrm{HH}^1(G, \mathrm{End}(E))$ classifies infinitesimal deformations of the twin system (E, α, G); (ii) $\mathrm{HH}^2(G, \mathrm{End}(E))$ contains the obstructions to extending first-order deformations; (iii) rigidity of the twin system under G-equivariant deformations is equivalent to the vanishing of HH^1; (iv) explicit computations are provided for cyclic, abelian, dihedral, and compact Lie groups, as well as matrix algebras; (v) a Hochschild–Serre spectral sequence is established.

This yields a deformation–obstruction theory for twin spaces that parallels, but is conceptually distinct from, classical Gerstenhaber deformation theory.

Contents

1 Introduction

Classical deformation theory investigates how algebraic structures vary when perturbed, and was developed in foundational works of Gerstenhaber [1], Nijenhuis–Richardson [2], Drinfeld [3], and Kontsevich [4]. A central theme is that infinitesimal deformations are controlled by cohomology groups. The prototype is the deformation theory of an associative algebra (A, μ): a first-order perturbation $\mu \mapsto \mu + t\,\mu_1$ remains associative to order t exactly when μ_1 is a Hochschild 2-cocycle, two such perturbations are equivalent exactly when they differ by a 2-coboundary, and the obstruction to integrating μ_1 to second order lives in $HH^3(A, A)$. Thus HH^2 classifies and HH^3 obstructs. The theory developed below follows the same script—cocycles classify, coboundaries trivialise, the next degree obstructs—but the object being deformed is not a single multiplication: it is an entire *orbit of operations* produced by a symmetry group, and the controlling cohomology is correspondingly different.

In Article I we introduced *twin operations*, a new algebraic construction attached to a group action. Let E be a set with a left action of a group G, and let $\alpha : E \times E \to E$ be a binary operation. Rather than focus only on G-invariant operations, we consider the entire orbit of α under the induced action

$$\alpha_g(x, y) := g^{-1} \cdot \alpha(g \cdot x,\ g \cdot y), \qquad g \in G. \tag{1.1}$$

The resulting family of operations

$$O = \{\alpha_g : g \in G\}$$

is not arbitrary: there is a canonical bijection

$$O \ \cong\ L\backslash G \qquad \text{(right cosets)}, \qquad |O| = [G : L],$$

where

$$L := \{g \in G : \alpha_g = \alpha\}$$

is the *linear subgroup*. We call this orbit of operations, equipped with the underlying action, a *twin space*.

Before developing any cohomology it is worth seeing the orbit O explicitly in a case where the operations genuinely move.

Example 1 (The translated additions)**.** Let $E = \mathbb{R}$, let $\alpha(x, y) = x + y$ be ordinary addition, and let $G = \mathrm{Aff}(\mathbb{R}) = \{g_{a,b} : x \mapsto ax + b \mid a \neq 0\}$ act on $\mathbb{R}$. Using $g_{a,b}^{-1}(u) = (u - b)/a$ we compute

$$x\,\alpha_{g_{a,b}}\,y \ =\ g_{a,b}^{-1}\big((ax + b) + (ay + b)\big) \ =\ \frac{a(x + y) + 2b - b}{a} \ =\ x + y + \tfrac{b}{a}.$$

So $\alpha_{g_{a,b}}$ is "addition shifted by $c = b/a$", and the twin orbit is the one-parameter family $O = \{(x, y) \mapsto x + y + c : c \in \mathbb{R}\}$. The linear subgroup is $L = \{g_{a,0}\}$, the pure scalings (for which $c = 0$), and indeed $O \cong L\backslash\mathrm{Aff}(\mathbb{R}) \cong \mathbb{R}$ through $g_{a,b} \mapsto b/a$. Each α_g is isomorphic to α

(it is carried onto α by g), yet the operations are pairwise distinct on the fixed set $\mathbb{R}$: a single operation has been spread by the symmetry into a whole line of operations. Deformation theory asks how such a family can be perturbed G-equivariantly, and which perturbations are genuinely new.

This viewpoint—from a single operation to a *space of operations generated by symmetry*—has three structural consequences that are central to this article:

(i) **New deformation complex.** The deformation problem concerns the entire twin space $O \cong L\backslash G$ rather than α alone. This leads naturally to a Hochschild-type cochain complex

$$C^\bullet(G, \mathrm{End}(E))$$

governing deformations of the twin structure (E, α, G).

(ii) **Geometric classification mechanism.** Because O is a homogeneous space $L\backslash G$, classifying compatible operations becomes a geometric and group–theoretic problem of describing subgroups of G containing L.

(iii) **Deformation–transport duality.** Deforming α means deforming the *transport rule*

$$\alpha \mapsto \alpha_g,$$

not the operation alone. Infinitesimal deformations correspond to infinitesimal variations in this transport, and are measured precisely by $\mathrm{HH}^1(G, \mathrm{End}(E))$.

The purpose of this article is to develop a full deformation cohomology theory for twin spaces: we construct a Hochschild-type complex, prove that HH^1 classifies infinitesimal deformations, identify obstruction classes in HH^2, provide cohomological rigidity criteria, and compute these groups in a range of examples, including a Hochschild–Serre spectral sequence of Hochschild–Serre type [5].

2 Twin Operations and Group Actions

Let G act on E from the left. Given a binary operation $\alpha : E \times E \to E$, we define

$$\alpha_g(x, y) := g^{-1} \cdot \alpha(g \cdot x,\ g \cdot y). \tag{2.1}$$

Definition 2 (Linear subgroup)**.** The *linear subgroup* of (E, α, G) is

$$L := \{g \in G : \alpha_g = \alpha\}.$$

Proposition 3. *The map*

$$L\backslash G \longrightarrow O, \qquad Lg \mapsto \alpha_g,$$

is a well-defined bijection. In particular, O is canonically isomorphic to the homogeneous space $L\backslash G$ of right *cosets, and $|O| = [G : L]$.*

Proof. Surjectivity is immediate from the definition $O = \{\alpha_g : g \in G\}$. For injectivity, suppose $\alpha_g = \alpha_h$. Then for all $x, y \in E$,

$$g^{-1} \cdot \alpha(g \cdot x, g \cdot y) = h^{-1} \cdot \alpha(h \cdot x, h \cdot y).$$

Applying g to both sides gives

$$\alpha(g \cdot x, g \cdot y) = gh^{-1} \cdot \alpha(h \cdot x, h \cdot y).$$

Substituting $(x, y) \mapsto (h^{-1} \cdot x, h^{-1} \cdot y)$ yields

$$\alpha(gh^{-1} \cdot x, gh^{-1} \cdot y) = gh^{-1} \cdot \alpha(x, y),$$

so $gh^{-1} \in L$, that is, $Lg = Lh$. Conversely, if $Lg = Lh$ then $gh^{-1} \in L$ and reversing the steps gives $\alpha_g = \alpha_h$; hence $Lg \mapsto \alpha_g$ is well defined and injective. □

Remark 4 (Right cosets and group structure). The identification is with the space of *right* cosets $L\backslash G$, not G/L: the assignment $g \mapsto \alpha_g$ is a *right* action, since a direct computation gives $\alpha_{hg} = (\alpha_h)_g$ — apply the defining formula twice, $(\alpha_h)_g(x, y) = g^{-1}h^{-1}\,\alpha(hg \cdot x,\, hg \cdot y) = \alpha_{hg}(x, y)$ — so the stabilizer of α is L and the orbit is $L\backslash G$ by orbit–stabilizer. Consequently $|O| = [G : L]$ unconditionally, and O is a homogeneous G-set. It carries a compatible group structure $O \cong G/L$ precisely when $L \trianglelefteq G$ (e.g. whenever G is abelian); in general L need not be normal. All cohomological results below depend only on the G-module structure of $M = \mathrm{End}(E)$ and are unaffected by this distinction.

This orbit description is the geometric backbone of twin spaces.

3 The Hochschild-Type Complex for Twin Operations

Set $M := \mathrm{End}(E)$ and let G act on M by

$$(g \cdot A)(x) := g \cdot A(g^{-1} \cdot x), \qquad A \in M,\ g \in G. \tag{3.1}$$

The action (3.1) is the natural *conjugation* action of G on linear endomorphisms: g transports a state back by g^{-1}, applies A, and transports the result forward by g. It makes $M = \mathrm{End}(E)$ a left G-module, and the complex defined below is precisely the inhomogeneous bar complex computing the group cohomology $H^\bullet(G, M)$. The deformation-theoretic dictionary is then transparent: a 1-cochain $f : G \to M$ records, for each symmetry g, the infinitesimal endomorphism by which the transported operation α_g is allowed to drift; the cocycle equation $\delta f = 0$ says these drifts are compatible with composition of symmetries; and the coboundaries are exactly the drifts induced by an infinitesimal change of coordinates on E. This is the “deformation–transport duality” announced in Section 1, and we make it precise in Section 4.

Definition 5 (Cochains)**.** For $n \geq 0$, the group of n-cochains is

$$C^n(G, M) := \{f : G^n \to M\}.$$

Definition 6 (Hochschild-type differential)**.** The differential $\delta : C^n(G, M) \to C^{n+1}(G, M)$ is given by

$$\begin{aligned}(\delta f)(g_1, \ldots, g_{n+1}) = {} & g_1 \cdot f(g_2, \ldots, g_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(g_1, \ldots, g_i g_{i+1}, \ldots, g_{n+1}) \\ & + (-1)^{n+1} f(g_1, \ldots, g_n).\end{aligned}$$

Proposition 7. $\delta^2 = 0$.

Proof. Write $\delta = \sum_{i=0}^{n+1} (-1)^i d_i$, where $d_0 f = g_1 \cdot f(g_2, \ldots, g_{n+1})$, the map d_i $(1 \leq i \leq n)$ contracts the adjacent pair $g_i g_{i+1}$, and d_{n+1} drops the last argument. These satisfy the simplicial identities $d_i d_j = d_{j-1} d_i$ for $i < j$: for the contraction maps they hold by associativity of the group law, and the two identities involving d_0 use the module axiom $g_1 \cdot (g_2 \cdot A) = (g_1 g_2) \cdot A$.

Substituting into $\delta^2 = \sum_{i,j}(-1)^{i+j} d_i d_j$ and re-indexing, the terms cancel in pairs. Each term of the form $g_1 g_2 \cdot f(\cdots)$ appears twice with opposite sign, and similarly for all intermediate combinations. One recovers exactly the usual cancellation in the bar complex for group cohomology. $\square$

Definition 8 (Twin deformation cohomology)**.** We define

$$\mathrm{HH}^n(G, M) := \frac{\mathrm{Ker}(\delta : C^n \to C^{n+1})}{\mathrm{Im}(\delta : C^{n-1} \to C^n)}.$$

This is the Hochschild-type cohomology governing deformations of the twin system (E, α, G).

4 Low-Degree Cohomology

We describe HH^0, HH^1, and HH^2 and their deformation-theoretic meaning.

4.1 HH^0: invariant endomorphisms

A 0-cochain is an element $A \in M$. Then

$$(\delta A)(g) = g \cdot A - A.$$

Thus

$$\mathrm{HH}^0(G, M) = M^G := \{A \in M : g \cdot A = A \ \forall g \in G\},$$

the subspace of endomorphisms commuting with the G-action.

4.2 HH^1: infinitesimal deformations

We consider a formal deformation

$$\alpha_t(x, y) = \alpha(x, y) + t\, \psi_1(x, y) + O(t^2).$$

Definition 9 (First-order G-equivariance)**.** We say that α_t is G-equivariant to first order if

$$g \cdot \alpha_t(x, y) - \alpha_t(g \cdot x, g \cdot y) = O(t^2)$$

for all $g \in G$.

Expanding in t, and using (2.1), we obtain

$$g \cdot \psi_1(x, y) - \psi_1(g \cdot x, g \cdot y) = 0,$$

which is precisely the cocycle condition $\delta f = 0$ for the 1-cochain $f \in C^1(G, M)$ that records the infinitesimal variation of the transport rule. (It is essential to keep the two roles distinct: ψ_1 perturbs the *operation* as a bilinear map, while f is its shadow on the *transport* $g \mapsto \alpha_g$, valued in $\mathrm{End}(E)$. It is the transport—equivalently the equivariance of the family $\{\alpha_g\}$—whose deformations are governed by $H^\bullet(G, \mathrm{End}(E))$, which is why the cochains take values in $\mathrm{End}(E)$ rather than in bilinear maps.)

Theorem 10. *Infinitesimal G-equivariant deformations of the twin system (E, α, G) are classified, up to equivalence, by* $\mathrm{HH}^1(G, M)$.

Proof. Let $f \in C^1(G, M)$ encode the infinitesimal change in the transport rule $\alpha \mapsto \alpha_g$. The first-order G-equivariance condition is exactly $\delta f = 0$. Two deformations differing by a change of variables $x \mapsto x + t\, \varphi(x)$ correspond to f and $f + \delta\varphi$, so they define the same class in HH^1. Thus equivalence classes of infinitesimal deformations are in bijection with $\mathrm{HH}^1(G, M)$. $\square$

4.3 HH^2: obstructions

To extend a first-order deformation to second order

$$\alpha_t = \alpha + t\psi_1 + t^2\psi_2 + O(t^3)$$

one obtains an equation of the form

$$\delta\psi_2 = \Theta(\psi_1)$$

for a certain 2-cochain $\Theta(\psi_1)$, determined quadratically from ψ_1. The consistency condition is that $\Theta(\psi_1)$ is a coboundary.

Theorem 11. *The obstruction to extending an infinitesimal deformation to second order is a cohomology class in* $\mathrm{HH}^2(G, M)$. *The deformation extends if and only if this class vanishes.*

Proof. The second-order G-equivariance condition has the form

$$g \cdot \alpha_t(x, y) - \alpha_t(g \cdot x, g \cdot y) = O(t^3),$$

which at order t^2 yields an equation

$$g \cdot \psi_2(x, y) - \psi_2(g \cdot x, g \cdot y) = \Xi(\psi_1)(g; x, y),$$

where the right-hand side is an explicit quadratic expression in ψ_1. At the cochain level this is $\delta\psi_2 = \Theta(\psi_1)$, for some $\Theta(\psi_1) \in C^2(G, M)$ with $\delta\Theta(\psi_1) = 0$. Hence $\Theta(\psi_1)$ defines a class $[\Theta(\psi_1)] \in \mathrm{HH}^2(G, M)$. A solution ψ_2 exists if and only if $\Theta(\psi_1)$ is a coboundary, i.e. $[\Theta(\psi_1)] = 0$. □

Remark 12 (The obstruction as a Maurer–Cartan square)*.* The quadratic obstruction $\Theta(\psi_1)$ is the cohomological avatar of the self-interaction of the first-order term. Equip the cochain spaces $C^\bullet(G, M)$ with the cup-type product inherited from composition in $M = \mathrm{End}(E)$ and the resulting graded bracket $[\,\cdot, \cdot\,]$; then $(C^\bullet(G, M), \delta, [\,\cdot, \cdot\,])$ is a differential graded Lie algebra, and a direct computation gives $\Theta(\psi_1) = \frac{1}{2}[f, f]$ for the classifying 1-cocycle f. The identity $\delta\Theta = 0$ is then the graded Jacobi relation, and the integrability of f to a full formal deformation is precisely the Maurer–Cartan equation $\delta\xi + \frac{1}{2}[\xi, \xi] = 0$ for $\xi = tf + t^2\psi_2 + \cdots$. This is the exact analogue, in the twin setting, of the role played by the Hochschild DGLA in associative-algebra deformation theory.

4.4 Rigidity

Definition 13 (Infinitesimal rigidity)**.** The twin system (E, α, G) is *infinitesimally rigid* if every infinitesimal G-equivariant deformation is equivalent to the trivial one.

Corollary 14. (E, α, G) *is infinitesimally rigid if and only if* $\mathrm{HH}^1(G, M) = 0$.

Proof. Immediate from Theorem 10. □

5 Deformation–Obstruction Theory for Twin Spaces

We now formulate the full deformation–obstruction framework.

Definition 15 (Formal G-equivariant deformation)**.** A *formal G-equivariant deformation* of α is a formal power series

$$\alpha_t(x, y) = \sum_{k \geq 0} t^k \alpha_k(x, y), \qquad \alpha_0 = \alpha,$$

such that for all $k \geq 0$,

$$g \cdot \alpha_t(x, y) - \alpha_t(g \cdot x, g \cdot y) = O(t^{k+1}).$$

Write $\alpha_t = \alpha + t\psi_1 + t^2\psi_2 + \cdots$. At each order we obtain

$$\delta\psi_k = \Theta_k(\psi_1, \dots, \psi_{k-1}),$$

where Θ_k is a universal expression, quadratic in the lower-order terms $\psi_1, \dots, \psi_{k-1}$ in the Gerstenhaber sense; in particular Θ_2 depends only on ψ_1, quadratically.

Theorem 16 (Deformation–obstruction correspondence)**.** *Let (E, α, G) be a twin system, with $M = \mathrm{End}(E)$.*

(i) Equivalence classes of infinitesimal G-equivariant deformations are in bijection with $\mathrm{HH}^1(G, M)$.

(ii) For each infinitesimal class $[f] \in \mathrm{HH}^1(G, M)$, the obstruction to extending it to a second-order deformation is a cohomology class in $\mathrm{HH}^2(G, M)$.

(iii) A complete formal deformation (to all orders) exists if and only if all the obstruction classes vanish in $\mathrm{HH}^2(G, M)$ at each stage.

Proof. Item (i) is Theorem 10; item (ii) is Theorem 11. For (iii), the higher-order extension steps give equations $\delta\psi_k = \Theta_k(\psi_1, \dots, \psi_{k-1})$ with $\delta\Theta_k = 0$. At each stage the obstruction is a class in $\mathrm{HH}^2(G, M)$, and the existence of a complete formal solution is equivalent to the vanishing of all of these classes. □

Definition 17 (Formal rigidity)**.** The twin system (E, α, G) is *formally rigid* if any formal G-equivariant deformation is equivalent to the trivial one.

Theorem 18 (Cohomological rigidity)**.** *If $\mathrm{HH}^1(G, M) = 0$ and $\mathrm{HH}^2(G, M) = 0$, then (E, α, G) is formally rigid.*

Proof. If $\mathrm{HH}^1(G, M) = 0$, every infinitesimal deformation is trivial. If in addition $\mathrm{HH}^2(G, M) = 0$, there are no non-trivial obstruction classes and spectral sequences of obstructions collapse trivially. An induction on the order in t shows that any putative deformation is equivalent to the trivial one. □

6 Explicit Cohomology Computations

We now compute $\mathrm{HH}^\bullet(G, M)$ in several important cases.

6.1 Finite cyclic groups

Let $G = C_n = \langle g \mid g^n = e \rangle$ act semisimply on a finite-dimensional k-vector space E. The action extends to $M = \mathrm{End}(E)$.

The bar resolution for C_n is 2-periodic and yields standard formulas in terms of the operators

$$(T-1) : M \to M, \quad A \mapsto g \cdot A - A,$$

and the norm

$$N : M \to M, \quad A \mapsto A + g \cdot A + \cdots + g^{n-1} \cdot A.$$

Under mild genericity/semisimplicity assumptions (no non-trivial invariants except M^G), one has:

Theorem 19. *Let $G = C_n$ act semisimply on E and $M = \mathrm{End}(E)$ as above. Then for all $k \geq 0$,*

$$\mathrm{HH}^{2k+1}(G, M) = 0, \qquad \mathrm{HH}^{2k}(G, M) \cong M^G.$$

In particular, $\mathrm{HH}^1(G, M) = 0$ and the corresponding twin system is infinitesimally rigid.

Proof. From the 2-periodic resolution, one has

$$H^{2k}(C_n, M) \cong \frac{\mathrm{Ker}(T-1)}{\mathrm{Im}(N)}, \qquad H^{2k+1}(C_n, M) \cong \frac{\mathrm{Ker}(N)}{\mathrm{Im}(T-1)}.$$

Semisimplicity implies the decomposition $M \simeq M^G \oplus M'$ with no invariants in M'. One checks that $\mathrm{Ker}(N) = \mathrm{Im}(T-1)$ and $\mathrm{Ker}(T-1) = M^G \oplus \mathrm{Im}(N)$, so odd cohomology vanishes and even cohomology identifies with M^G. Since our deformation cohomology $\mathrm{HH}^\bullet(G, M)$ coincides with this group cohomology, the result follows. □

Example 20 (A rigid involution)**.** Let $\mathrm{char}\, k \neq 2$, $E = k^2$, and $G = C_2 = \{e, \sigma\}$ with σ the coordinate swap $(x_1, x_2) \mapsto (x_2, x_1)$. On $M = \mathrm{End}(E) \cong M_2(k)$ the group acts by conjugation by the swap matrix $S = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, namely $\sigma \cdot A = SAS^{-1}$. The invariants $M^G = \{A : SAS^{-1} = A\}$ are the matrices commuting with S, i.e. the 2-dimensional centraliser $\{aI + bS : a, b \in k\}$. Since $\mathrm{char}\, k \neq 2$ the action is semisimple, so Theorem 19 applies and gives

$$\mathrm{HH}^{2k}(C_2, M) \cong M^G \ (\dim 2), \qquad \mathrm{HH}^{2k+1}(C_2, M) = 0.$$

In particular $\mathrm{HH}^1 = 0$: the associated twin system is infinitesimally rigid. The two-dimensional even cohomology is detected by the invariant endomorphisms I and S themselves.

6.2 Finite abelian groups

Let G be finite abelian and act semisimply on E and $M = \mathrm{End}(E)$. We have a decomposition

$$G \cong \mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_r\mathbb{Z}.$$

Using Künneth, one obtains:

Theorem 21. *Let G be a finite abelian group acting semisimply on E. Then for all $k \geq 0$,*

$$\mathrm{HH}^k(G, M) \cong \bigwedge^k \mathrm{Hom}(G, k) \otimes M^G.$$

In particular,

$$\mathrm{HH}^1(G, M) \cong \mathrm{Hom}(G, k) \otimes M^G.$$

Proof. The cohomology $H^\bullet(G, k)$ is known to be an exterior algebra on $\mathrm{Hom}(G, k)$, and semisimplicity together with the universal coefficient theorem yields the stated tensor product with M^G. □

6.3 Compact Lie groups

Let G be a compact connected Lie group acting smoothly on a smooth manifold E. Consider $M = C^\infty(E)$ with the natural action $g \cdot f(x) = f(g^{-1} \cdot x)$.

Theorem 22. *If G is compact connected and acts smoothly on E, then for all $k \geq 1$,*

$$\mathrm{HH}^k(G, C^\infty(E)) = 0.$$

Proof sketch. One uses Haar integration to define an averaging operator on cochains:

$$(Af)(g_1, \dots, g_n)(x) = \int_G f(hg_1h^{-1}, \dots, hg_nh^{-1})(h^{-1} \cdot x)\, d\mu(h),$$

which projects onto conjugation-invariant cochains. For compact connected groups, such invariant cochains in the smooth setting are coboundaries in positive degrees, by the vanishing of suitable continuous group cohomology and the connectedness of G. Thus $\mathrm{HH}^k(G, C^\infty(E))$ vanishes for $k \geq 1$. □

6.4 Matrix algebras and conjugation

Let $E = M_n(k)$ and $\alpha(X, Y) = XY$ be matrix multiplication. Let $G = GL_n(k)$ act by conjugation

$$g \cdot X = gXg^{-1}.$$

Then $M = \mathrm{End}(E)$ carries a highly semisimple representation.

Theorem 23. *In the above setting,*

$$\mathrm{HH}^1(G, M) = 0, \qquad \mathrm{HH}^2(G, M) = 0.$$

Hence the matrix twin system is formally rigid.

Proof sketch. Over a field of characteristic zero the group GL_n is *linearly reductive*: every rational representation splits as a direct sum of irreducibles, and the invariants functor $W \mapsto W^{GL_n}$ is exact (it is split by the Reynolds operator). An exact invariants functor has no higher derived functors, so the rational cohomology of GL_n vanishes in positive degree: $H^i(GL_n, W) = 0$ for every rational module W and every $i \geq 1$. Now $M_n(k) \cong V \otimes V^*$ with $V = k^n$ under the conjugation action, so $M = \mathrm{End}(M_n(k))$ is again a rational GL_n-module; taking rational (polynomial) cochains—the natural choice for an algebraic group action—the vanishing applies to $W = M$ and yields $H^1 = H^2 = 0$, hence HH^1 and HH^2 in our setting. □

7 Spectral Sequences for Twin Deformation Cohomology

Let $1 \to N \to G \to Q \to 1$ be a short exact sequence of groups. The action of G on M restricts to N and descends to Q.

Theorem 24 (Hochschild–Serre spectral sequence)**.** *There is a spectral sequence*

$$E_2^{p,q} = \mathrm{HH}^p(Q, \mathrm{HH}^q(N, M)) \Longrightarrow \mathrm{HH}^{p+q}(G, M).$$

Proof sketch. Filter the bar resolution of G by the number of arguments in N and Q, obtaining a double complex whose total complex computes $\mathrm{HH}^\bullet(G, M)$. The associated spectral sequence has the indicated E_2-page, as in [5]. Convergence follows from standard homological algebra arguments. □

Example 25 (Dihedral groups)**.** For the dihedral group D_{2n} with normal cyclic subgroup generated by the rotation, the spectral sequence

$$E_2^{p,q} = \mathrm{HH}^p(\mathbb{Z}/2\mathbb{Z}, \mathrm{HH}^q(\mathbb{Z}/n\mathbb{Z}, M))$$

allows explicit computations of $\mathrm{HH}^\bullet(D_{2n}, M)$ using the results of Theorems 19 and 21.

8 Rigidity Criteria for Twin Systems

We summarize the cohomological criteria for rigidity.

Theorem 26 (Rigidity via HH^1 and HH^2)**.** *Let (E, α, G) be a twin system with $M = \mathrm{End}(E)$.*

(i) (E, α, G) is infinitesimally rigid if and only if $\mathrm{HH}^1(G, M) = 0$.

(ii) If $\mathrm{HH}^1(G, M) = 0$ and $\mathrm{HH}^2(G, M) = 0$, then (E, α, G) is formally rigid.

Proof. Item (i) is Corollary 14. Item (ii) is Theorem 18. □

Corollary 27 (Compact connected groups)**.** *In the setting of Theorem 22, the associated twin system is formally rigid.*

Corollary 28 (Matrix twin systems)**.** *In the setting of Theorem 23, the matrix twin system is formally rigid.*

9 Examples and Deformation Spectra

We illustrate the variety of deformation behaviors.

Example 29 (Rigid cyclic rotations)**.** Let $E = \mathbb{R}^d$ and $G = C_n$ act by a generic orthogonal rotation with no non-trivial fixed subspace except $\{0\}$. Then by Theorem 19, $\mathrm{HH}^1(G, M) = 0$ and the twin system is infinitesimally rigid.

Example 30 (A flexible free-abelian example)**.** Take $G = \mathbb{Z} = \langle g \rangle$ acting on $E = k^2$ by $g = \mathrm{diag}(1, \mu)$ with $\mu \neq 0, 1$, so that the conjugation action on $M = \mathrm{End}(E) \cong M_2(k)$ is semisimple. Group cohomology of $\mathbb{Z}$ is computed by the length-one free resolution $0 \to \mathbb{Z}[\mathbb{Z}] \xrightarrow{g-1} \mathbb{Z}[\mathbb{Z}] \to \mathbb{Z} \to 0$, which gives $\mathrm{HH}^0 = \ker(g - 1) = M^G$, $\mathrm{HH}^1 = \mathrm{coker}(g - 1) = M/(g-1)M$, and $\mathrm{HH}^{\geq 2} = 0$. The invariants M^G are the matrices commuting with $\mathrm{diag}(1, \mu)$, namely the diagonal matrices, of dimension 2; by semisimplicity $M = M^G \oplus (g - 1)M$, so $\mathrm{coker}(g - 1) \cong M^G$ and hence $\mathrm{HH}^1 \cong M^G$ is also 2-dimensional. Thus the twin system is genuinely *flexible*: it carries a 2-parameter family of pairwise inequivalent infinitesimal deformations, in agreement with the exterior-algebra computation underlying Theorem 21 applied to the free abelian group $\mathbb{Z}$ (for which $\mathrm{Hom}(\mathbb{Z}, k) = k$). A *finite* abelian group over a characteristic-zero field would instead have $\mathrm{Hom}(G, k) = 0$ and hence $\mathrm{HH}^1 = 0$; the flexibility here comes precisely from the infinite, free direction of $\mathbb{Z}$.

Example 31 (Dihedral twin systems)**.** For D_{2n} acting on vertices or edges of a regular n-gon, the spectral sequence of Theorem 24 combined with Theorem 19 describes $\mathrm{HH}^\bullet(D_{2n}, M)$, typically leading to non-trivial HH^1 but constrained higher-degree cohomology.

Example 32 (Compact actions on spheres)**.** Let $G = SO(n)$ act on $E = S^{n-1}$. In many natural smooth settings, Theorem 22 implies vanishing of positive cohomology degrees, and hence rigidity of rotationally symmetric twin operations on the sphere.

10 Conclusion and Outlook

In this article we have developed a full deformation cohomology theory for *twin spaces*, i.e. triples (E, α, G) equipped with the orbit of twin operations

$$\alpha_g(x, y) = g^{-1} \cdot \alpha(g \cdot x, g \cdot y), \qquad g \in G,$$

and the identification $O \cong L\backslash G$ (right cosets) with the linear subgroup L. The key idea is that the relevant object is not the operation α alone, but its entire orbit of operations generated by the group action. This orbit structure is the conceptual basis for attaching a Hochschild-type cohomology complex $C^\bullet(G, \mathrm{End}(E))$.

We have:

- constructed the complex $C^\bullet(G, \mathrm{End}(E))$ and defined $\mathrm{HH}^\bullet(G, \mathrm{End}(E))$;
- identified $\mathrm{HH}^0(G, \mathrm{End}(E))$ with invariant endomorphisms;
- shown that $\mathrm{HH}^1(G, \mathrm{End}(E))$ classifies infinitesimal G-equivariant deformations of the twin space;
- identified obstruction classes in $\mathrm{HH}^2(G, \mathrm{End}(E))$;
- established cohomological criteria for infinitesimal and formal rigidity;
- computed $\mathrm{HH}^\bullet(G, \mathrm{End}(E))$ in several families (finite cyclic and abelian groups, dihedral groups, compact Lie groups, matrix algebras);
- constructed a Hochschild–Serre spectral sequence for twin deformation cohomology.

Conceptually, twin spaces provide a bridge between symmetry, geometry and deformation: the group action organizes operations into a homogeneous orbit $L\backslash G$, and the way this space deforms is precisely encoded by the cohomology $\mathrm{HH}^\bullet(G, \mathrm{End}(E))$. In other words, the twin formalism is not a cosmetic reformulation; it is the structural reason why a rich deformation–obstruction theory exists in this setting.

Possible directions for future work include:

(a) **Categorical and operadic refinements.** In companion papers we promote twin spaces to objects of a 2-category endowed with monoidal and operadic structure. Understanding how $\mathrm{HH}^\bullet$ interacts with these higher structures is a natural next step.

(b) **Quantum and Hopf-theoretic applications.** Twin spaces can be used to reinterpret certain quantum deformations of algebras as deformations of transport rules on twin operations, suggesting a new viewpoint on quantum groups and Hopf algebras.

(c) **Derived and homotopical generalizations.** Replacing sets and vector spaces by chain complexes or higher stacks should lead to derived versions of twin spaces, with deformation controlled by higher or ∞-categorical analogues of $\mathrm{HH}^\bullet(G, \mathrm{End}(E))$.

(d) **Geometric and analytic implementations.** For actions on manifolds or analytic spaces, twin spaces offer a framework in which geometric constraints (compactness, regularity, curvature) translate directly into cohomological constraints on the admissible deformations.

Altogether, twin spaces emerge as a robust and flexible foundation for encoding how operations transform under symmetry, with a deformation theory rich enough to connect to classical and modern themes in algebra, geometry, and mathematical physics.

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